

Cybersecurity

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# Project Introduction and Linux Hardening

Lesson 10.1



Welcome



Let's Recap



# Recap

Throughout this course, you've learned:

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- 1 Terminal
- 2 Systems administration
- 3 Networking

# Project 1 Overview

Project week will run a little differently. Each day will proceed with:

01

A brief lecture introducing new concepts.

02

A brief overview of project tasks.

03

A provided guide to help you work on completing your tasks.



## Important



Everyone will submit their own project. You can choose to work independently or in groups, but are **required to remain in class**.

Each day's lesson builds on the previous day, so all activities are **required to be completed in order**.

# Daily Objectives and Milestones

## Day 1

Harden an organization's linux server's users, groups, files, and directories.

## Day 2

Configure the SSH settings, system packages, services, and logging configurations.

## Day 3

Automate and schedule tasks from Day 1 and Day 2.

# Project Deliverables

As you work through the project, you will develop the following deliverables:

Completed script

Checklist and project  
summary





# Learning Objectives

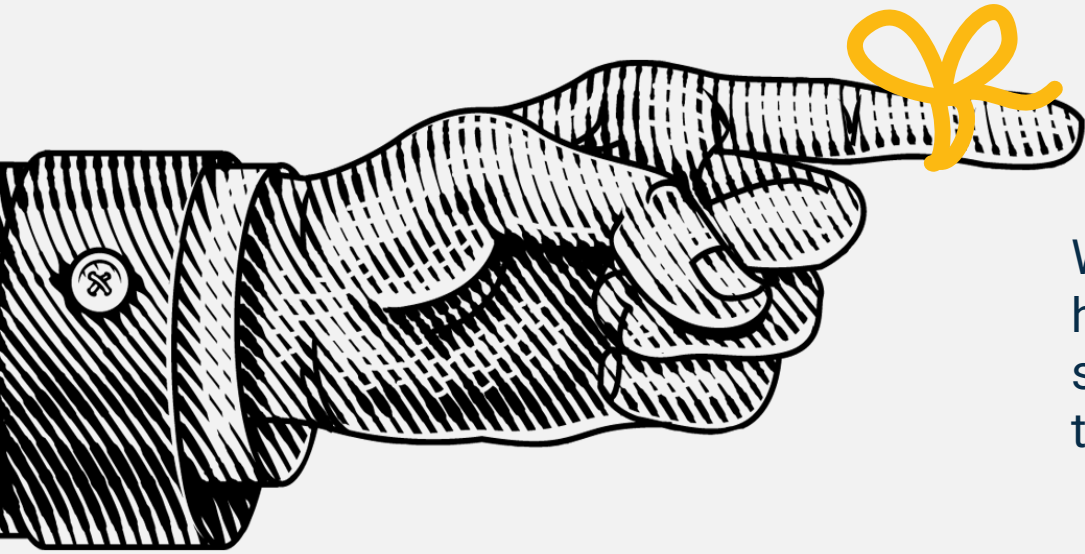
Today's lesson will cover:

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- 1 Introduction to Linux hardening.
- 2 Overview of tasks to complete on Day 1.
- 3 Project work.

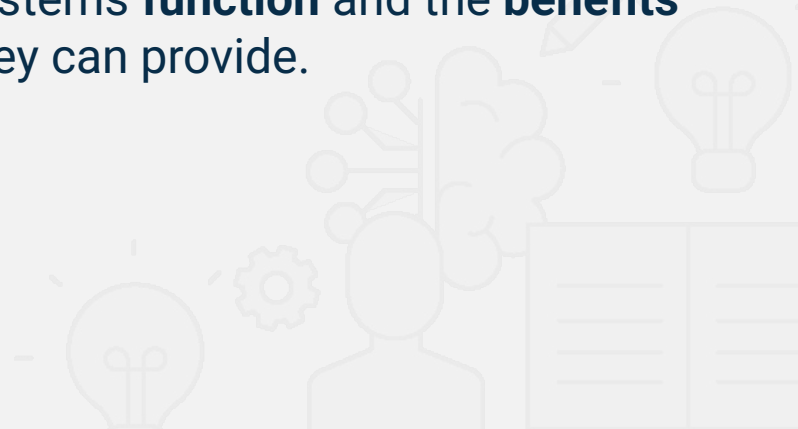


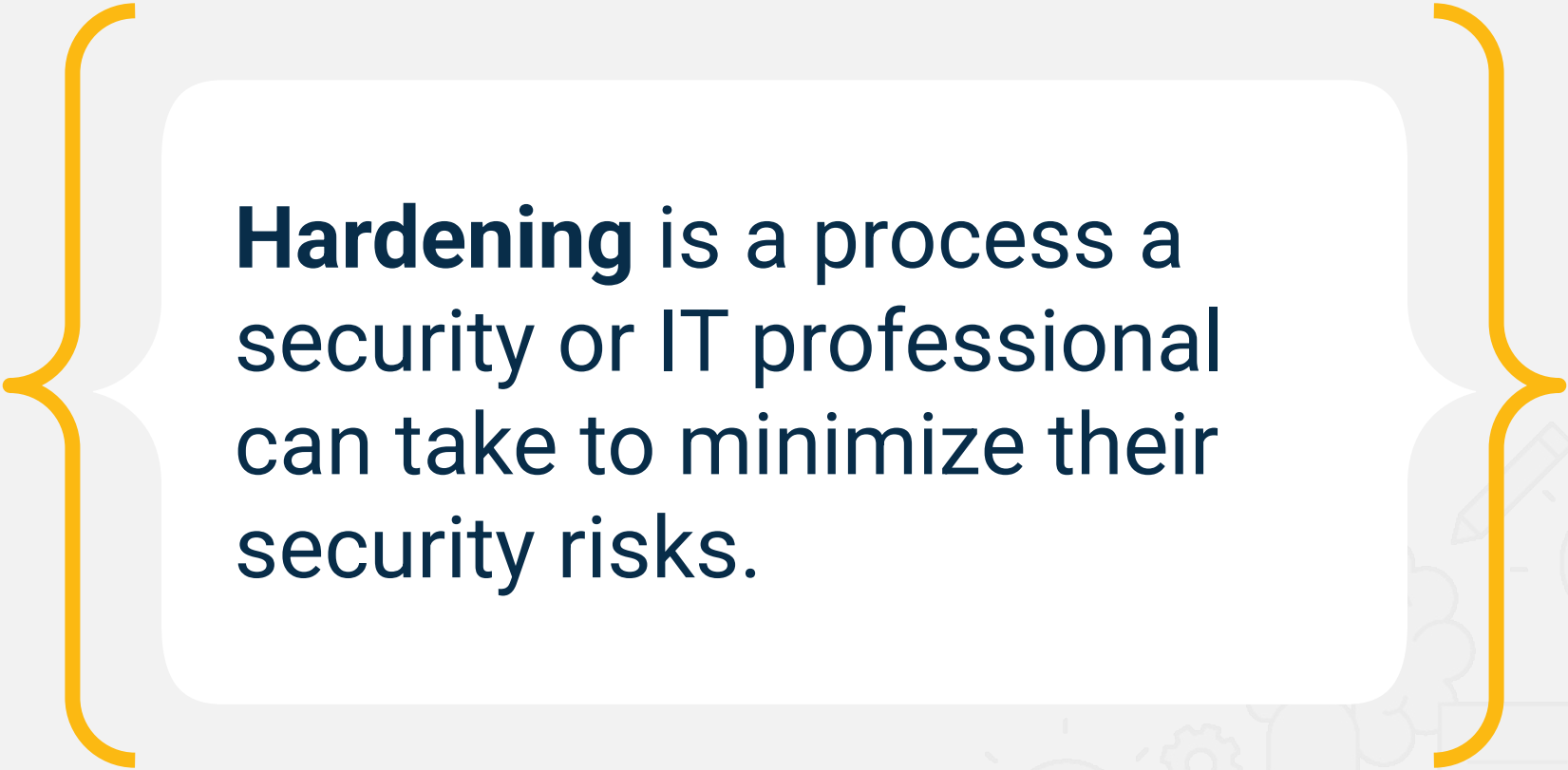
# Hardening Your **Linux** Server



# Remember

We have had several units that have illustrated how Linux operating systems **function** and the **benefits** they can provide.





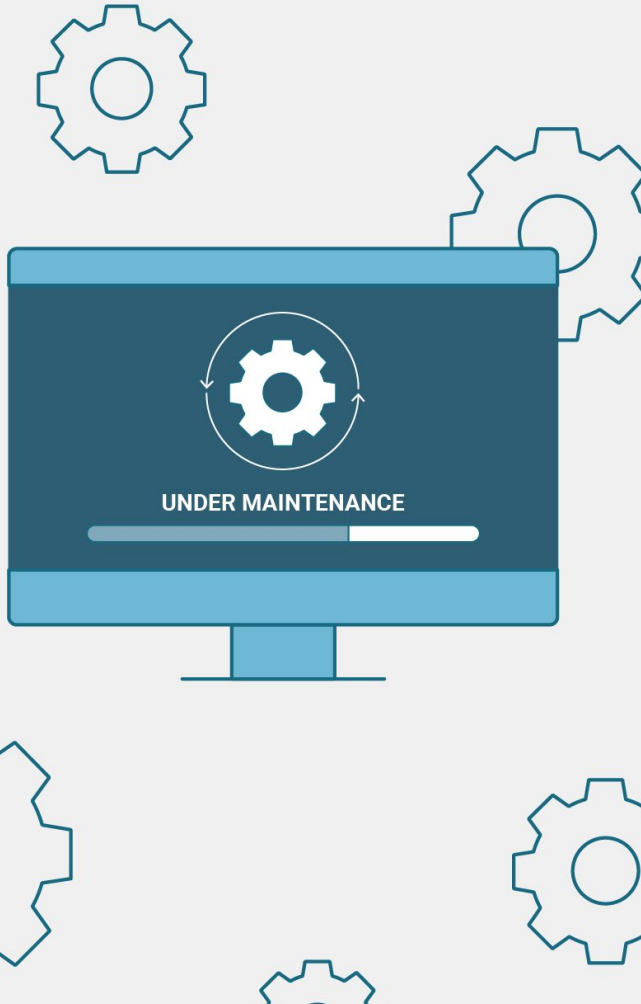
**Hardening** is a process a security or IT professional can take to minimize their security risks.

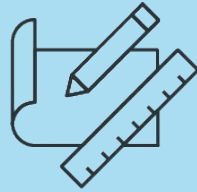


# Linux Hardening

Hardening steps can include:

- Assuring correct permissioning and system access.
- Removing unnecessary services, processes, and applications.
- Utilizing secure configuration options.
- Patching and updating current systems and processes.





# Day 1 **Project Overview**

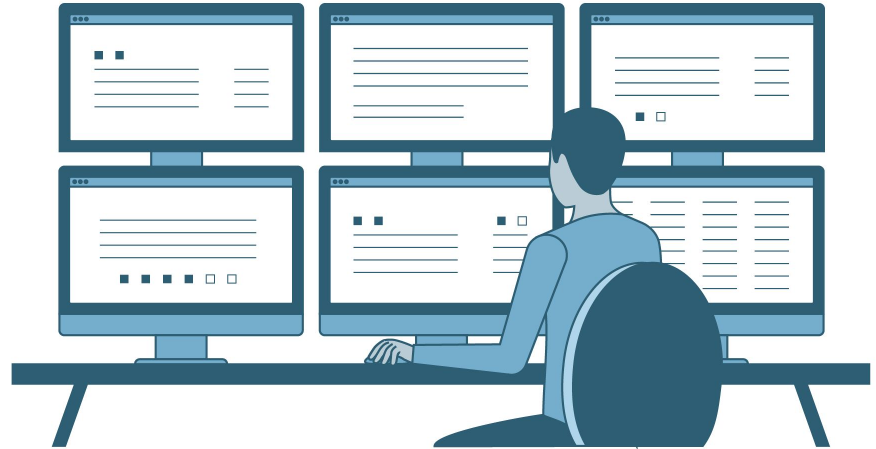


# Project Scenario

You will play the role of a security professional tasked with hardening a Linux server owned by the firm The Baker Street Corporation (BSC).

BSC has **confidential data** on their server, therefore they need you to confirm their system is properly configured to protect them from security breaches.

If you identify any security issues, they are also requesting that you make the necessary updates.



# Project Documentation

You will be provided with two documents:

## Guide

Assist with the steps for Day 1:

- System inventory and backup
- Auditing users and groups
- Password policies
- `sudo` permissions
- File and directory permissions

## Hardening Summary

Checklist document that you will be submitting as a deliverable at the end of your project.

**Remember** to update the checklist as you progress through each activity.



# Day 1 Project Overview

Pre-hardening steps: System inventory and backup

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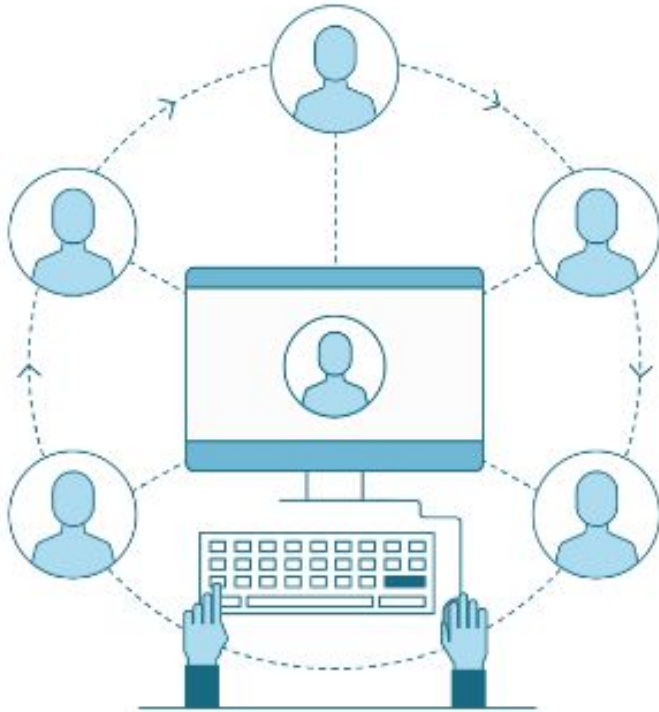
- Document details about the operating system.
- Create a backup.





# Day 1 Project Overview

## Auditing users and groups

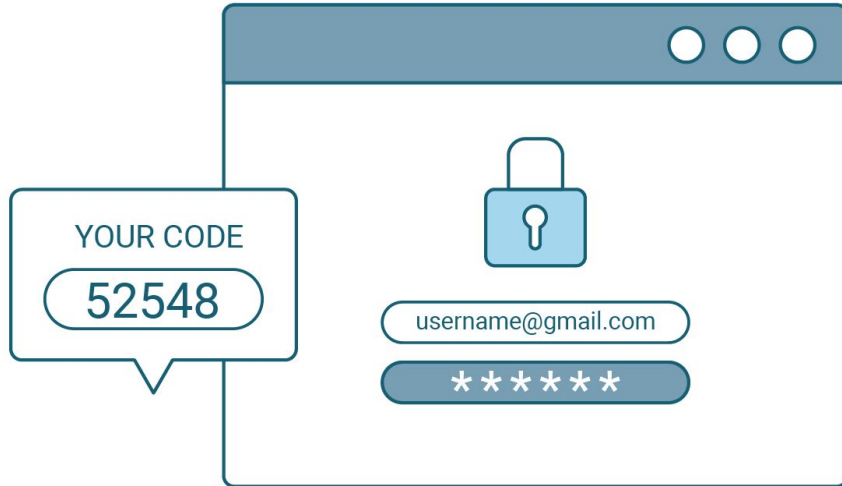


- Ensure that only the correct BSC employees and groups have the minimal access needed to do their jobs.
- Check and update the security permissions of BSC employees and groups.



# Day 1 Project Overview

## Updating and enforcing password policies



- Identify BSC employees with weak passwords.
- Create and implement strict password policies.
- Force users to update passwords.

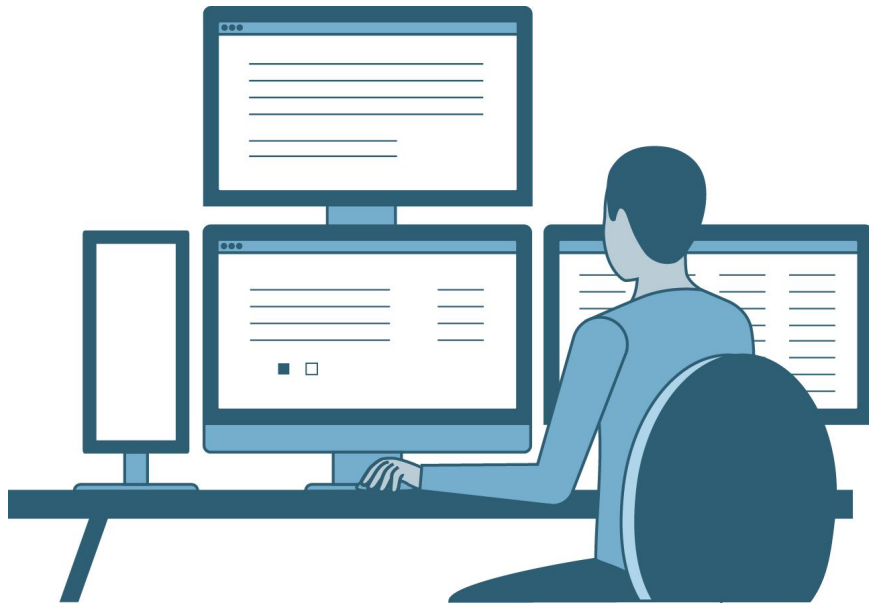


# Day 1 Project Overview

Updating and enforcing `sudo` permissions

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- Validate and update the `sudoers` file.
- Only grant `sudo` privileges for specific tasks.



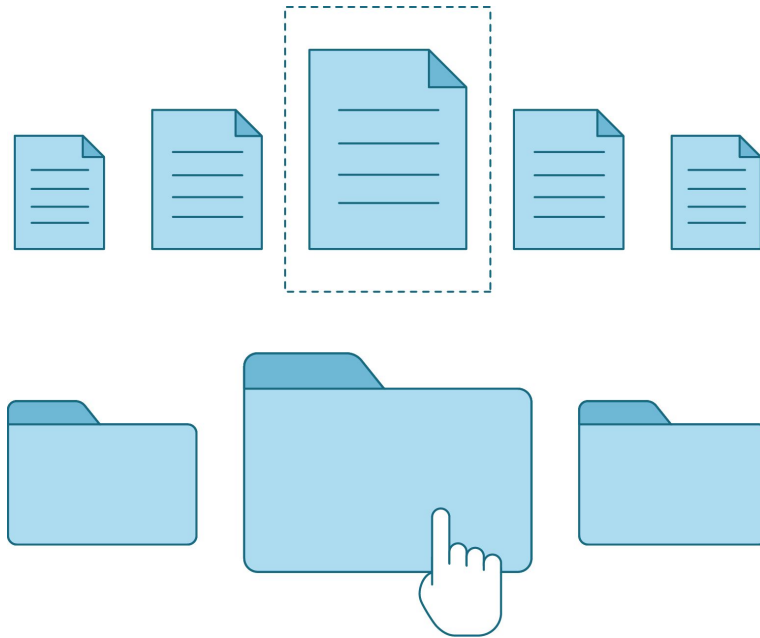


# Day 1 Project Overview

Validating and updating permissions on files and directories

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- Check files and directories.
- Fix any security issues you discover.





Questions?





# Activity

## Hardening your Linux server

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For the remainder of today's class, you will work on completing Day 1 project tasks.

- While you are responsible for completing your own project, you can ask classmates, the TA, or the instructor to assist if you have any questions.
- **Remember** to complete the checklist document as you progress through the activities.

**Suggested Time**  
Remainder of class



Time's Up!  
Let's review



Questions?





# Next Lesson

Day 2: Hardening your Linux server's system settings



Thank You